

AMIT SINGH RAJAWAT

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in amit-singh

amitsingh-24

Website

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, JavaScript, Shell, MySQL
- **Developer Tools & Frameworks:** Git, GitHub, Linux, Docker, ONNX, AWS, CI/CD, MLOps
- **Libraries:** PyTorch, TensorFlow, Scikit-learn, Flask, OpenCV, Transformers, MXNet, LLMs, FastAPI

WORK EXPERIENCE

Software Engineer

June 2024 – Present

Kloudspot Inc.

Bengaluru, India

- Devised a **Face Recognition Algorithm** in **C++** for **1:1 Verification** and **1:N Identification**, achieving a global rank of **169** in **1:1 Verification** and **59** in **1:N Identification** (NIST-Face Recognition Vendor Test – FRVT).
- Built a **Face Recognition System (FRS)** Analysis Tool using **clustering techniques** and an **FRS Registration Tool** in **Python**, implementing detection, landmark processing, and recognition for **multi-pose registration** with **real-time tracking** and **visualization**.
- Implemented a custom **NVIDIA DeepStream** parser for **face detection** and **landmark extraction**, generating optimized **engine files** for **NVIDIA platforms** to enable **high-performance, real-time facial analysis** in **video processing pipelines**.
- Engineered **scalable data pipelines** for **real-time facial recognition**, integrating **CI/CD practices** for **automated deployment** and **continuous improvement of deep learning models**, while collaborating with product managers and analytics teams to define **performance metrics**, conduct **exploratory data analysis**, and optimize **ML solutions** for enhanced **operational efficiency** and **business impact**.
- Trained models using **multiple architectures** (ResNet, ResNet with CBAM, ResNet with attention, etc.), incorporating rigorous **hyperparameter tuning**, **knowledge distillation**, and **reinforcement learning** techniques to enhance model performance and improve **accuracy by 13%**.

Software Engineering Intern

Feb 2024 – May 2024

Kloudspot Inc.

Bengaluru, India

- Built a **high-performance** computer vision inferencing framework in **C++** using **GStreamer**, **OpenVINO**, and **DLStreamer**, resulting in a **9% improvement** in **processing speed**.
- Architected and implemented a **gRPC-based GStreamer RTSP Server** for **efficient client-server communication**, optimizing **pre-processing** and **post-processing workflows** to enhance **model prediction accuracy** and **system efficiency** for **large-scale data handling**, aligning with **real-time solutions**.

Machine Learning Intern (LOR)

Dec 2023 – Jan 2024

Indian Space Research Organisation (ISRO)

Bengaluru, India

- Designed a **vision-based satellite pose estimation system** using the **HRNet algorithm**, **iterative hyperparameter tuning**, and **real-time processing optimizations** to achieve **90% accuracy** in **landmark detection** across **diverse and challenging imaging conditions**.
- Integrated **deep landmark regression** and **nonlinear pose refinement** techniques, reducing **error rates by 15%** through **extensive model training**.

PROJECTS

EchoNeuron AI Engine | Python, FastAPI, LangChain, ChatGroq, Tavily, JavaScript, Render

(Link)

- Built a **unified assistant** featuring **live web lookup**, **DuckDuckGo search**, **ArXiv search**, **YouTube** and **PDF summarization**, **tutoring**, and **quiz generation** using **FastAPI**, **LangChain**, and **ChatGroq**.
- Created a **GUI** containerized with **Docker** and implemented a **secure CI/CD pipeline** on **Render** with safe handling of **Groq** and **Tavily API keys**.

StoryCraftAI - Infinite Possibilities in Every Story | Python, TensorFlow, Keras, NLP, Pandas, Huggingface, CI/CD

(Link)

- Developed and optimized a suite of **deep learning models** for **story generation** using **advanced NLP techniques**, improving model accuracies: **GRU** increased by **13.46%**, **LSTM** by **33.53%**, **Bidirectional-LSTM** by **25.32%**, and **Bidirectional-GRU** by **9.91%**.
- Integrated an **interactive Flask-based web interface** for **real-time story generation** and deployed the solution via **CI/CD** on **Hugging Face Spaces**.

GestureSymphony - Control Video Playback Using Your Hands | Python, OpenCV, MediaPipe, Pygame, Flask, NumPy, Bootstrap

(Link)

- Created a **real-time hand gesture recognition system** using **MediaPipe Hands** and **OpenCV** for **detection and control**, demonstrating strong **computer vision** and **ML integration**.
- Enhanced **system efficiency** by implementing **lazy loading** and containerizing with **Docker**; established a **robust CI/CD pipeline** for **continuous deployment** on **cloud platforms** like **Render**.

EDUCATION

Madhav Institute of Technology & Science, Gwalior

Aug 2020 – May 2024

B.Tech in Information Technology (Artificial Intelligence & Robotics)

CGPA: 8.65

PUBLICATIONS

- A. Rajawat, A. Manjhi and A. Srivastava, "Categorization of IRIS Flower using Different Machine Learning Strategies," [paper]
- A. Manjhi, A. Rajawat and A. Srivastava, "Design and Analysis of Programmed Face Monitoring System," [paper]
- Rajawat, A.S., Srivastava, A., "Recognition of Parkinson's Ailment by Using Various Machine Learning Procedures." Current Psychology Journal (2024). [paper]
- **On Heart Disease** (Communicated in Journal).

ACHIEVEMENTS

- **Leetcode:** **990+ Problems Solved** & **1765+ Rating** (Profile Link), **GFG:** **320+ Problems Solved** (Profile Link).
- Selected for **IIT Madras Pravartak Undergraduate Fellowship 2023-24** (among **72 in India**) (Link).
- Led the development of the **NEC Portal** for the college, currently used by over **5,000 students** (Link).
- Led a **3-day hackathon** at **MITS**, orchestrating **team coordination** and contributing to the core **organizing committee**.
- Selected for the **Amazon ML Summer School**, implementing **machine-learning algorithms** in **Python** to achieve over **95% accuracy** and building **sequence-modeling** and **Spark-based pipelines** for **high-throughput, low-latency inference**.
- **Certifications:** Advanced Learning Algorithms, DeepLearning.AI TensorFlow Developer Specialization, Machine Learning.